

Tutorail-7

- ✓ Q1. Let $ALL_{DFA} = \{ \langle A \rangle \mid A \text{ is a DFA and } L(A) = \Sigma^* \}$. Show that ALL_{DFA} is decidable.
- ✓ Q2. Let $A_{ECFG} = \{ \langle G \rangle \mid G \text{ is a CFG that generates } \epsilon \}$. Show that A_{ECFG} is decidable.
- ✓ Q3. Let B be the set of all infinite sequences over $\{0, 1\}$. Show that B is uncountable, using a proof by diagonalization.
- ✓ Q4. Let $T = \{ (i, j, k) \mid i, j, k \in \mathbb{N} \}$. Show that T is countable.
- ✓ Q6. Let $INFINITE_{DFA} = \{ \langle A \rangle \mid A \text{ is a DFA and } L(A) \text{ is an infinite language} \}$. Show that $INFINITE_{DFA}$ is decidable.
- ✓ Q7. Let $INFINITE_{PDA} = \{ \langle M \rangle \mid M \text{ is a PDA and } L(M) \text{ is an infinite language} \}$. Show that $INFINITE_{PDA}$ is decidable.
- ✓ Q8. Let $A = \{ \langle M \rangle \mid M \text{ is a DFA which doesn't accept any string containing an odd number of 1s} \}$. Show that A is decidable.
- ✓ Q9. Let $A = \{ \langle R, S \rangle \mid R \text{ and } S \text{ are regular expressions and } L(R) \subseteq L(S) \}$. Show that A is decidable.
- ✓ Q10. Let $Z = \{0, 1\}$. Show that the problem of determining whether a CFG generates some string in 1^* is decidable.
- ✓ Q11. Show that the problem of determining whether a CFG generates all strings in 1^* is decidable.
- ✓ Q12. Let $A = \{ \langle R \rangle \mid R \text{ is a regular expression describing a language containing at least}$

one string w that has 111 as a substring (i.e., $w = x111y$ for some x and y). Show that A is decidable.

Q13. Prove that EQ_{DFA} is decidable by testing the two DFAs on all strings up to a certain size (called limit DFA).

Q14. Let C be a language. Prove that C is Turing-recognizable iff a decidable language D exists such that $C = \{x \mid \exists y (\langle x, y \rangle \in D)\}$.

Q15. Let $S = \{\langle M \rangle \mid M \text{ is a DFA that accepts } w^R \text{ whenever it accepts } w\}$. Show that S is decidable.

Q16. A language is prefix-free if no member is a proper prefix of another member. Let $PREFIX-FREE_{REG} = \{R \mid R \text{ is a regular expression where } L(R) \text{ is prefix-free}\}$. Show that $PREFIX-FREE_{REG}$ is decidable.

Q17. Say that an NFA is ambiguous if it accepts some string along two different computation branches. Let $AMBIG_{NFA} = \{\langle N \rangle \mid N \text{ is an ambiguous NFA}\}$. Show that $AMBIG_{NFA}$ is decidable.

Q18. A useless state in a pushdown automaton is never entered on any input string. Consider the problem of determining whether a pushdown automaton has any useless states. Formulate this problem as a language and show that it is decidable.

Q19. Repeat Q18 the same for DFA.

Q20. Repeat Q18 the same for NFA.

Q21. Let $BAL_{DFA} = \{\langle M \rangle \mid M \text{ is a DFA that accepts some string containing an equal number of 0s and 1s}\}$. Show that BAL_{DFA} is decidable.

✓ Q22. Let $BALN_{DFA} = \{ \langle M \rangle \mid M \text{ is a DFA that accepts some string containing a non-equal number of 0s and 1s} \}$. Is that $BALN_{DFA}$ decidable? Comment on closure of TMs on implementation.

✓ Q23. Let $PAL_{DFA} = \{ \langle M \rangle \mid M \text{ is a DFA that accepts some palindrome} \}$. Show that PAL_{DFA} is decidable.

✓ Q24. Let $E = \{ \langle M \rangle \mid M \text{ is a DFA that accepts some string with more 1s than 0s} \}$. Show that E is decidable.

✓ Q25. Let $C = \{ \langle G, x \rangle \mid G \text{ is a CFG that generates some string } w, \text{ where } x \text{ is a substring of } w \}$. Show that C is decidable.